

1 January 8, 2026 (Class 0)

SCS CVX Julia DCP - disciplined convex programming SOCP
Norm approximation

2 January 9, 2026 (Class 1)

2.1 Norm Approximation

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^m$. The objective is

$$\min_{\mathbf{x} \in \mathbb{R}^n} \|\mathbf{Ax} - \mathbf{b}\|$$

Definition 2.1 (ℓ_1 norm): For $\mathbf{y} \in \mathbb{R}^m$, ℓ_1 norm is defined as

$$\|\mathbf{y}\|_1 = \sum_{i=1}^m |y_i|$$

Definition 2.2 (ℓ_2 norm): For $\mathbf{y} \in \mathbb{R}^m$, ℓ_2 norm is defined as

$$\|\mathbf{y}\|_2 = \left(\sum_{i=1}^m y_i^2 \right)^{1/2}$$

Definition 2.3 (ℓ_∞ norm): For $\mathbf{y} \in \mathbb{R}^m$, ℓ_∞ norm is defined as

$$\|\mathbf{y}\|_\infty = \max_{1 \leq i \leq m} |y_i|$$

3 January 14, 2026 (Class 2)

3.1 Penalty Functions

3.1.1 Norm Based ($p = 1, 2, \infty$)

$$\|\cdot\|_1, \|\cdot\|_2, \|\cdot\|_\infty.$$

3.1.2 Dead Zone Linear

Functions of the form, for $r \in \mathbb{R}^n$,

$$\sum_i \phi(r_i),$$

where

$$\phi(r) = \max\{|r| - a, 0\}.$$

3.1.3 Log Barrier

$$\phi(u) = \begin{cases} -\log(1 - u^2), & |u| < 1 \\ \infty, & |u| \geq 1 \end{cases}$$

3.1.4 Huber Penalty

3.2 Moreau Envelope

Definition 3.1 (Moreau Envelope): Given $f : \mathbb{R}^n \rightarrow \mathbb{R}$, we have the Moreau Envelope as

$$f_u(\mathbf{x}) = \min_{\mathbf{y} \in \mathbb{R}^n} \left\{ f(\mathbf{y}) + \frac{1}{2u} \|\mathbf{x} - \mathbf{y}\|^2 \right\}.$$

3.2.1 Properties of Moreau Envelope

1. $f_u(\mathbf{x})$ is differentiable if f is continuous. Further, the derivative is continuous.
2. $f_u(\mathbf{x})$ is defined for all $\mathbf{x} \in \mathbb{R}^n$, even when f is not.
3. $f_u(\mathbf{x})$ is convex whenever f is convex.
4. If x^* is the unique minimizer of f , then it is the unique minimizer of $f_u(\mathbf{x})$.

4 January 16, 2026 (Class 3)

Remark 4.1: $f_u(\mathbf{x})$ is convex, since it is the min over one variable of a jointly convex function.

Remark 4.2: $f_u(\mathbf{x})$ is continuously smooth.

Remark 4.3:

$$\text{epi} f_u(x) = \text{epi} f(x) \oplus \text{epi} \left(\frac{1}{2u} \|x\|^2 \right)$$

Remark 4.4: x^* is a minima of f if and only if x^* is a minima of f_u .

Remark 4.5:

$$\nabla f_u(\mathbf{x}) = \frac{1}{u} (1 - \text{prox}_u(x)),$$

where,

$$\text{prox}_u(x) = \arg \min_{\mathbf{y} \in \mathbb{R}^n} \left\{ f(\mathbf{y}) + \frac{1}{2u} \|\mathbf{x} - \mathbf{y}\|^2 \right\}.$$

5 January 20, 2026 (Class 4)